

STA 640 — Causal Inference

Chapter 2.2. Randomized Experiments: Covariate Balance and Adjustment

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Covariates in Randomized Experiments

- ▶ Randomized experiments or randomized controlled trials (RCTs)
- ▶ Randomization “should” balance all baseline covariates (observed and unobserved) **on average...**
- ▶ But covariates may be imbalanced by random chance, and sometimes better balance is desired
- ▶ Why is covariate balance important? Because better balance
 - ▶ Provides more meaningful estimates of the causal effect – internal validity
 - ▶ increases precision (reduces variance) of estimator, if covariates correlated with outcome (outcome less variable for similar values of covariates)

Covariate Balance: Mean Standardized difference (SMD)

- ▶ In a causal study, a good practice is always to first check covariate balance
- ▶ Many metrics of balance; a traditional one is Mean Standardized Difference (SMD)

$$d_1 = \left| \frac{\sum_{i=1}^N X_i Z_i}{\sum_{i=1}^N Z_i} - \frac{\sum_{i=1}^N X_i (1 - Z_i)}{\sum_{i=1}^N (1 - Z_i)} \right| / \sqrt{s_1^2/N_1 + s_0^2/N_0},$$

s_z^2 : sample variance of covariate in group z for $z = 0, 1$

- ▶ d_1 : dependent on sample size, same mean difference, larger sample size results in larger d_1
- ▶ For a continuous covariate, d_1 is the standard two-sample t-statistic, and the threshold is based on a t- or z- test (e.g. 1.96)

Covariate Balance: Mean Standardized Difference (SMD)

- ▶ Debate on whether N_0 and N_1 should be in the denominator: with large sample size, imbalance will always be declared based on d_1
- ▶ In clinical research, a popular alternative SMD drops N in denominator:

$$d_2 = \left| \frac{\sum_{i=1}^N X_i Z_i}{\sum_{i=1}^N Z_i} - \frac{\sum_{i=1}^N X_i (1 - Z_i)}{\sum_{i=1}^N (1 - Z_i)} \right| / \sqrt{(s_1^2 + s_0^2)/2},$$

- ▶ d_2 is scale-free, a special case of Cohen's d (effect size).
- ▶ **The common threshold of d_2 is 0.1** (Austin, 2011), ad hoc because d_2 no underlying distribution, but popular
- ▶ More general, multivariate, balance metrics are available: e.g.
(1) variance ratios, (2) Kolmogorov-Smirnov (KS) statistic
- ▶ Always good to check higher order terms and interactions

Example: BestAIR trial

Zeng et al., 2021 SIM

TABLE 2 Baseline characteristics of the BestAIR randomized trial by treatment groups, and absolute standardized difference (ASD) between the treatment and control groups before and after weighting

	All patients N = 169	CPAP group N ₁ = 83	Control group N ₀ = 86	ASD ^{UNADJ}	ASD ^{IPW}	ASD ^{OW}
Baseline categorical covariates and number of units in each group						
Gender (male)	107	54	53	0.046	0.002	0.000
Race and ethnicity						
White	152	75	77	0.051	0.015	0.000
Black	11	5	6	0.060	0.007	0.000
Other	5	2	3	0.086	0.034	0.000
Recruiting center						
Site 1	54	26	28	0.046	0.002	0.000
Site 2	10	5	5	0.065	0.024	0.000
Site 3	105	52	53	0.073	0.013	0.000
Baseline continuous covariates, mean and standard deviation (in parenthesis)						
Age (years)	64.4 (7.4)	64.4 (8.0)	64.3 (6.8)	0.020	0.017	0.000
BMI (kg/m ²)	31.7 (6.0)	31.0 (5.3)	32.4 (6.5)	0.261	0.042	0.000
Baseline SBP (mm Hg)	124.3 (13.2)	121.6 (11.1)	127.0 (14.6)	0.477	0.020	0.000
Baseline SDP (beats/min)	63.1 (10.7)	63.0 (10.4)	63.2 (10.9)	0.020	0.016	0.000
Baseline AHI (events/h)	28.8 (15.4)	26.5 (13.0)	31.1 (17.2)	0.348	0.039	0.000
Baseline ESS	8.3 (4.5)	8.0 (4.5)	8.5 (4.6)	0.092	0.010	0.000

Note: The ASD^{OW} is exactly zero due to the exact balance property of OW.

Covariate adjustment: BestAIR trial

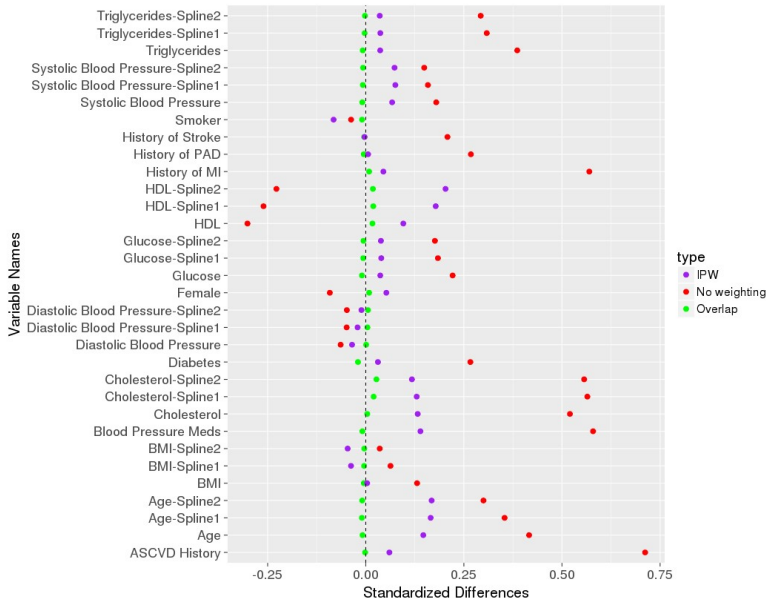
Zeng et al., 2021 SIM

- ▶ The Best Apnea Interventions for Research (BestAIR) trial: individually-randomized clinical trial on patients with high cardiovascular disease risk and obstructive sleep apnea
- ▶ Goal: evaluate the effect of continuous positive airway pressure (CPAP) treatment on health outcomes
- ▶ Treatment arm: 83 patients; control arm: 86 patients
- ▶ Outcome: 24-hour systolic blood pressure (SBP); self-reported sleepiness in daytime measured by Epworth Sleepiness Scale (ESS); both measured at 6 months
- ▶ Covariates: Patient-level characteristics, and outcomes measured at baseline (usually the most important covariate)
- ▶ Marked imbalance in the baseline covariates in BestAIR

Report and Visualize Balance

- ▶ In randomized trials, good balance is expected in all covariates, but not always, especially in small studies (see BestAir example)
- ▶ In observational studies, imbalance in covariates is expected (see Framingham study)
- ▶ Covariate-by-covariate before-after table of baseline characteristics, known as Table 1 in medical papers
- ▶ **Boxplot** of SMDs of important covariates, including higher order terms and splines
- ▶ **Love plot** (named about Thomas Love) of SMDs

Love Plot of SMDs in Framingham Study



Balance by Design in RCTs: Stratified Experiments

- ▶ Units are stratified (grouped, blocked) according to covariate(s)
- ▶ Subdivide sample into J homogeneous strata (blocks)
- ▶ Randomize units to treatment groups within each strata
- ▶ Often used with important categorical covariates (or discretized quantitative)
- ▶ (similar to stratified sampling)

Stratified Experiments: Bee Stings Example

Free, J.B. (1961) "The stinging response of honeybees," *Animal Behavior*, vol. 9, pp 193-196

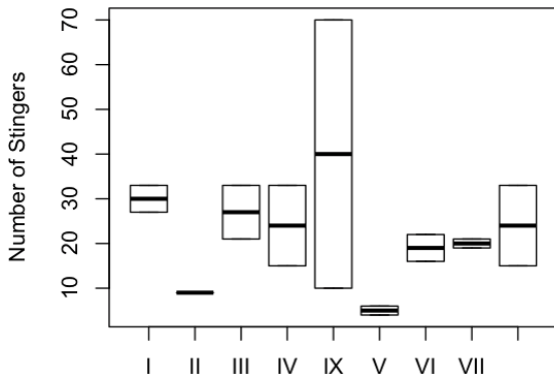
- ▶ If you are stung by a bee, does that make you more likely to get stung again? (Might bees leave behind a chemical message that tells other bees to attack you?)
- ▶ Scientists dangled 16 muslin-wrapped cotton balls over a beehive, where half of the balls had been previously stung and the other half were fresh
- ▶ Outcome: total number of new stingers
- ▶ Repeated for a total of nine trials

Covariate Balance

- ▶ Option 1: force better balance on important covariates *by design*
 - ▶ stratified randomized experiments
 - ▶ paired randomized experiments
 - ▶ rerandomization
- ▶ Option 2: correct imbalance in covariates by *analysis* – covariate adjustment
 - ▶ outcome: gain scores
 - ▶ separate analysis within subgroups
 - ▶ regression (ANCOVA)
 - ▶ propensity score weighting
 - ▶ model-based imputation

Stratified Experiments: Bee Stings Example

- ▶ Scientists expect the number of stings to vary by trial (different number of bees in the hive, different times of day, different weather, etc.)
- ▶ Each trial is a different strata ($J = 9$)



Stratified Experiments: Fisherian inference

- ▶ In Fisher's mode (randomization) of inference, what to use for a test statistic?
- ▶ A common choice: weighted average of the strata-specific differences

$$T = \sum_{j=1}^J \frac{N_j}{N} \left(\bar{Y}_{1,j}^{obs} - \bar{Y}_{0,j}^{obs} \right)$$

where $\bar{Y}_{z,j}^{obs}$ is the average outcome of group z in J th strata, N_j is the size of J th strata

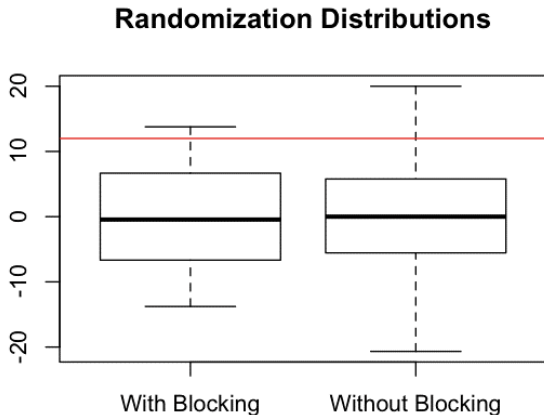
- ▶ Note: another common choice of weights by the variance of each strata – minimize the total variance of T

Stratified Experiments: Fisherian inference

- ▶ Fisher randomization test
- ▶ Easier to simulate randomizations, rather than enumerate all possible allocations
- ▶ For each simulated randomization, just randomize within strata
- ▶ Always for randomization test, just simulate randomization schemes compatible with the randomization design – think about what is the randomization space $\mathcal{S}$ in stratified experiments

Bee Stings Example: Fisher's test

- ▶ All trials are equal size (N_j constant), $T^{obs} = 12$
- ▶ p-value = 0.04



Stratified Experiments: Neymanian inference

- ▶ In Neyman's mode (repeated sampling) of inference, what to use for an estimator?
- ▶ In general, $\hat{\tau} = \sum_j \lambda_j \hat{\tau}_j$, where $\hat{\tau}_j$ and λ_j is the estimator and weight of strata j , respectively
- ▶ Variance of the general estimator $\hat{\tau}$ is $\mathbb{V}(\hat{\tau}) = \sum_j \lambda_j^2 \mathbb{V}(\hat{\tau}_j)$
- ▶ One common choice is $\hat{\tau}^{ave} = \sum_{j=1}^J \frac{N_j}{N} \left(\bar{Y}_{1,j}^{obs} - \bar{Y}_{0,j}^{obs} \right)$
- ▶ For the specific estimator $\hat{\tau}^{ave}$:

$$\mathbb{V}(\hat{\tau}^{ave}) = \sum_j \frac{N_j^2}{N^2} \left(\frac{s_{1,j}^2}{N_{1,j}} + \frac{s_{0,j}^2}{N_{0,j}} \right)$$

Stratified Experiments

- ▶ “Block what you can; randomize what you cannot” - George Box
- ▶ No harm – can only help
- ▶ Can stratify on more than one covariate
- ▶ Strata can be any size - more balanced design (i.e. similar strata size and similar sizes of treatment and control per strata) has more power
- ▶ Difficult to extend to many covariates or continuous covariates

Paired Experiments

- ▶ Units are matched into pairs (based on covariate(s))
- ▶ Special case of stratified randomized experiment with $N_j = 2$ for each strata
- ▶ Useful when expect difference in potential outcomes within a pair to be much smaller than differences across pairs
- ▶ Test statistic /estimator: average of differences across all pairs (note: this is the same as the difference of the averages within treatment groups)

Paired Experiments

- ▶ Randomization test: randomize sign of each difference
- ▶ Neyman inference: analyze differences as a single variable
- ▶ When variability within pairs is much smaller than variability across pairs, can get huge gains in precision
- ▶ Translate to higher power for tests (lower p-values) and narrower confidence intervals

Chance Imbalance in Randomized Experiments

- ▶ Randomization balances all covariates only *in expectation*, not necessarily in each single trial, particularly when the sample size is small
- ▶ Chance imbalance in baseline covariates commonly exists
- ▶ Some baseline covariates can be predictive of the outcome. For example, in randomized clinical trials, such a covariate is the baseline outcome.
- ▶ Chance imbalance in covariates that are predictive of the outcome can
 - ▶ reduce the power of the unadjusted estimator $\hat{\tau}^{\text{unadj}}$ detecting true difference if there is
 - ▶ threat the *face validity* of an experiment

Balance by Analysis: Covariate Adjustment

- ▶ Solution to chance imbalance: covariate adjustment
 - ▶ Regression adjustment
 - ▶ Propensity score weighting (later)
 - ▶ Model-based imputation/Bayesian (later)
- ▶ Goal: improve the precision (power) in estimating the causal effects

Unadjusted Estimator: Revisited

- ▶ Under randomization: $\tau = \mathbb{E}(Y_i|Z_i = 1) - \mathbb{E}(Y_i|Z_i = 0)$
- ▶ Unadjusted (diff-in-means) estimator $\hat{\tau}_{\text{unadj}}$, unbiased for τ :

$$\hat{\tau}_{\text{unadj}} = \bar{Y}_1 - \bar{Y}_0 = \frac{\sum_{i=1}^N Z_i Y_i}{\sum_{i=1}^N Z_i} - \frac{\sum_{i=1}^N (1 - Z_i) Y_i}{\sum_{i=1}^N (1 - Z_i)}$$

- ▶ $\hat{\tau}_{\text{unadj}}$ is equivalent to coefficient of Z of linear regression

$$\text{lm}(Y_i \sim 1 + Z_i)$$

- ▶ Adjusting for imbalanced covariates can reduce variance of $\hat{\tau}_{\text{unadj}}$
(Fisher, 1935)

Covariate Adjustment: ANCOVA

- ▶ Regression adjustment: fit a regression model of the observed Y on the treatment indicator Z , covariates X , and possibly interactions ZX ; use the ordinary least square (OLS) estimator of the coefficient of Z as the adjusted estimator of ATE τ
- ▶ Also known as ANCOVA estimator
- ▶ Two different ANCOVA models:
 - ▶ ANCOVA 1 (without interactions): $\text{lm}(Y_i \sim Z_i + X_i)$
 - ▶ ANCOVA 2 (with interactions): $\text{lm}(Y_i \sim Z_i + X_i + Z_i X_i)$
- ▶ In both models, the OLS estimator of the coefficient of Z , β_z , is taken as the estimator of τ , often denoted as the adjusted estimator $\hat{\tau}^{adj}$
- ▶ Question: Any difference between the two models? Which one is better?

ANCOVA Adjustment: Key Points

Lin, 2013 AOAS

- ▶ To obtain a ANCOVA estimator that is more efficient than the unadjusted estimator of the ATE τ in randomized experiments, one must
 1. include a full set of treatment-covariates interaction, that is, use ANCOVA2
 2. center X (so that you can directly use the OLS estimate of coefficient of Z in the ANCOVA as the adjusted estimator for τ)

Why do interactions matter?

- ▶ For simplicity, consider a single continuous covariate X
- ▶ Fit a regression model of the potential outcomes in treatment and control group separately

$$Y_i(1) = \alpha_1 + \beta_1 X_i + \epsilon_{i1}$$

$$Y_i(0) = \alpha_0 + \beta_0 X_i + \epsilon_{i0}$$

- ▶ Use the equation $Y = Y_i(1)Z_i + Y_i(0)(1 - Z_i)$ to connect the models on the potential outcomes and the observed outcome

$$\begin{aligned} Y &= (\alpha_1 + \beta_1 X_i)Z_i + (\alpha_0 + \beta_0 X_i)(1 - Z_i) + \epsilon_i \\ &= \alpha_0 + (\alpha_1 - \alpha_0)Z_i + \beta_0 X_i + (\beta_1 - \beta_0)X_i Z_i + \epsilon_i \end{aligned}$$

(1)

OLS Covariate-Adjusted Estimator

- ▶ In practice, we can first fit a linear model, $\text{lm}(Y \sim \tilde{X})$, separately in each group, and let $\hat{\beta}_1$ and $\hat{\beta}_0$ be the OLS estimate of the coefficient of X in these two regressions

- ▶ The covariate-adjusted estimator is:

$$\hat{\tau}^{\text{adj}} = \left\{ \bar{Y}_1 - \hat{\beta}'_1 (\bar{X}_1 - \bar{X}) \right\} - \left\{ \bar{Y}_0 - \hat{\beta}'_0 (\bar{X}_0 - \bar{X}) \right\}$$

- ▶ Numerically, $\hat{\tau}^{\text{adj}}$ is equivalent to the OLS estimate of the coefficient of Z in the following model with interactions with **centered** $\tilde{X}$

$$\text{lm}(Y \sim Z + \tilde{X} + Z\tilde{X})$$

- ▶ Asymptotically, $\hat{\tau}^{\text{adj}}$ (i.e. ANCOVA2 estimator) is more efficient than $\hat{\tau}^{\text{unadj}}$; in fact it is the most efficient estimator among the semiparametric family of covariate adjustment estimators (Tsiatis

Model-free validity for model-based statistics

- ▶ $\hat{\tau}^{\text{adj}}$ is biased for τ in finite samples (the bias is usually small and negligible), but is unbiased asymptotically (consistent)
- ▶ **Model-free validity for model-based statistics:** Consistency of $\hat{\tau}^{\text{adj}}$ holds even if the linear regression assumption is wrong
- ▶ Intuition
 - ▶ By randomization, the super-population correlation between the treatment indicator and covariates is zero
 - ▶ The correlation vanishes in large sample, but may not in finite sample
- ▶ Only holds in randomization - not true in observational studies!

OLS Estimators: Variance

Lin, 2013 AOAS

- ▶ To estimate the s.e. of $\widehat{\tau}^{\text{adj}}$ under randomization, use the **Huber–White sandwich s.e.** (also known as the heteroscedasticity-consistent estimator)

$$\widehat{\mathbb{V}}_{HW}(\widehat{\tau}^{\text{adj}}) = (X'X)^{-1}(X'\text{diag}(\hat{u}_1^2, \dots, \hat{u}_n^2)X)(X'X)^{-1},$$

where $\hat{u}_i$ are the regression residuals

- ▶ **Huber–White s.e.** is justified by randomization, but the traditional OLS s.e. is not
- ▶ H-W variance: `vcovHC` function from the `sandwich` R package, with several types (HC2 or HC3 often better than the original HC0). See Ding's causal book for code

Covariate adjustment in practice: Gap

- Barrier in practice: missing data in both outcomes and covariates (BestAir example)

	A	B	C	D	E	F	G	H	I	J	K	L
1	treatment	gender	age	race	bmi	sbp_base	dbp_base	ah1_base	ess_base	siteid	sbp_6mo	ess_6mo
2	1	1	49	1	27.48635	118.9264	75.82412	15.6	1	3	118.8582	1
3	1	2	70	1	46.68761	132.8707	61.26506	48.7	3	3	134.1148	2
4	1	1	68	1	35.12955	123.528	67.73014	19.5	10	1	125.77	9
5	1	1	67	1	32.42475	132.1843	75.91092	15.6	7	3	129.8734	11
6	1	1	54	1		119.2102	73.03658	19.2	12	1	125.9953	7
7	1	1	68	3	24.61726	111.525	69.28315	26.5	10	2	128.8968	8
8	1	1	55	1		157.4346	82.20229	18.8	5	3		
9	0	2	67	1	27.93019	131.2686	76.53263	15.8	4	3	153.025	6
10	0	1	76	1	35.40899	143.3476	64.22976	26.7	9	3	149.3891	
11	0	1	68	1	31.81545	131.0057	82.91501	23.1	11	3	151.0994	17
12	0	2	55	1	41.81889	102.6575	67.3125	28.6	6	1	103.1763	6
13	0	1	65	1	32.78341	119.7719	74.74392	17.6	12	1		12
14	0	2	55	1		156.8765	70.87031	33.4	8	1		
15	0	2	70	1	34.87619	122.5226	64.50238	21.6	16	3	147.1261	14
16	1	1	67	1	34.54368	135.2869	66.22847	25.7	11	2	131.2571	5
17	1	2	67	1	36.88581	117.5071	72.91369	9.62	2	3	116.5021	3
18	0	2	62	1	32.02981	148.0823	78.66171	19.2	11	3		12
19	1	1	74	1	21.9038	117.1287	55.39494	17.5	7	3	109.5813	3
20	1	2	63	1	25.63195	125.4815	66.80556	16.6	7	1	128.1157	4
21	0	1	60	1	28.88181	132.7219	69.239	40.2	6	3	132.0315	5
22	0	2	54	1	20.18684			18	7	3		6
23	0	1	48	1	32.57786	122.9262	78.35476	30.3	8	1		9
24	0	1	74	1	22.32093			18.5	10	3		10
25	0	1	69	1	27.08526	109.6825	65.08381	15.1	16	2	115.0929	11

- Covariate adjustment is a **secondary** analysis: Going extra mile to deal with missing data. How? Worth it?

RCT Covariate Adjustment: Missing data

- ▶ When there are missing data in the covariates: adjust or not?
how to adjust? impute the missing data? does the way of imputation matter?
- ▶ Zhao and Ding (2024): theoretically proved **specific imputation values/method do not matter, just use mean imputation and include the missingness indicators as predictors; $\hat{\tau}^{adj}$ is still asymptotically more efficient than the unadjusted estimator**
 - ▶ Assuming only one partially observed covariate X
 - ▶ R_i : the missingness indicator with $R_i = 1$ if X_i observed, $R_i = 0$ if X_i missing
 - ▶ The OLS model for covariate adjustment:

$$Y_i \sim Z_i + X_i R_i^x + R_i^x + Z_i (X_i R_i^x + R_i^x)$$

Covariate Adjustment in RCT with Missing Covariates

- ▶ Key insight: $X_i R_i^x$ is numerically equivalent to X_i when the covariate is observed and zero when the covariate is missing regardless how the missing value is imputed.
- ▶ Zhao and Ding (2024) result is on missing covariates only, regardless of the missing data mechanism of X (MAR, MCAR, MNAR).
- ▶ The result echoed and explained the previous results based on simulations
- ▶ In practice, most time there is missingness in both covariate and outcomes
- ▶ **Missing outcome**: different from **missing covariates**. Imputation quality matters a lot to the final conclusions (Zhao et al. 2024)

Covariate Adjustment in High Dimensions

- ▶ Recap: In randomized trials, covariate adjustment via interacted OLS regression can reduce variance of ATE (Lin, 2013)
- ▶ What if the covariates are high dimensional?
- ▶ Bloniarz et al. (2016): replace OLS with LASSO
 - ▶ Use post-LASSO (Belloni and Chernozhukov, 2013), i.e. first use LASSO for variable selection, and then perform OLS on the selected variable
 - ▶ Post-LASSO is performed separately in control and treatment group
- ▶ The LASSO-adjusted regression estimator is consistent and more efficient than the unadjusted difference-in-means ATE estimator

OLS Adjustment: History

- ▶ Classical idea dating back to Laplace (cf. Cochran 1977) - Laplace was, among many other things, the first (Bayesian) statistician
- ▶ Fisher (1935) proposed ANCOVA estimator in randomized experiments, argued that $\hat{\tau}^{\text{adj}}$ improves the precision (variance) over the unadjusted estimator $\hat{\tau}^{\text{unadj}}$
- ▶ Freedman (2008a,b) provided an antagonist view: gave examples where $\hat{\tau}^{\text{adj}}$ decrease precision (**unbalanced** experiments with **heterogeneity**, non-continuous outcome)
- ▶ Lin (2013) provided an agnostic view: re-examined Freedman's critique and provided simple fixes – case closed
- ▶ Tsiatis and co-authors made similar findings to Lin earlier (Yang and Tsiatis, 2001; Tsiatis et al., 2008)

Regression Adjustment: Limitations

- ▶ Limitations of regression adjustment in randomized experiments
 - ▶ Invitation for a *fishing expedition*: search for model specifications that give ‘significant’ result
 - ▶ Linear regression has nice properties justified by randomization, but other models (e.g., Logit or Probit) do **not** (Freedman 2008b)
 - ▶ Non-continuous (binary or count) outcomes: cannot directly use the coefficient of treatment in a linear model as the causal estimate
 - ▶ Rare non-continuous outcomes: a GLM may fail to converge
 - ▶ Some purists just disdain the invasion of a model(!) in the sacred temple of randomization
- ▶ Propensity score weighting methods can bypass these issues (later)
- ▶ All belong to the same semiparametric class of estimators (Tsiatis et al. 2008)

Words of Wisdom on OLS Adjustment

One does not need to believe in the classical linear model to tolerate or even advocate OLS adjustment, just as one does not need to believe in the Four Noble Truths of Buddhism to entertain the hypothesis that mindfulness meditation has causal effects on mental health.

— Winston Lin, *Agnostic notes on regression adjustments to experimental data: Reexamining Freedman's critique* (AOAS 2013, p 314)

Summary: Covariates in Randomized Experiments

- ▶ In randomized experiments, covariates may be included in the analysis for the following reasons:
 - ▶ the assignment mechanism depends on covariates (in a known way): e.g., block randomized experiments, cluster randomized experiments
 - ▶ covariates are useful in model-based inference for improving the prediction of the missing outcomes
- ▶ My experience: covariate adjustment in practice rarely changes the conclusion of the unadjusted estimator, therefore usually only reported in secondary analysis
- ▶ Randomization ensures *model-free validity for model-based statistics*

Important topics not covered

- ▶ Rerandomization
- ▶ Multi-arm (more than 2) experiments (later)
- ▶ Generalizability of random experiments: internal versus external validity (later)
- ▶ Noncompliance (later)
- ▶ interference

The following are additional slides for your interest (won't be covered in lectures)

Additional Slides on Cluster Randomized Experiments

- ▶ Cluster Randomized Experiments
 - ▶ treatment is assigned at cluster level but outcomes are measured for individuals within clusters
 - ▶ individuals in the same cluster receives the same treatment
 - ▶ also known as cluster randomized trials (CRT) or group randomized trials (GRT)
- ▶ Justified when
 - ▶ treatment is naturally delivered at the cluster level, e.g. villages, classrooms, clinics
 - ▶ prevent treatment contamination within clusters
 - ▶ ethical reasons

Estimand: Average Treatment Effect (CRT)

- ▶ Individual j ($j = 1, \dots, m_i$) in cluster i ($i = 1, \dots, n$) has two P.O's, $Y_{ij}(1)$ and $Y_{ij}(0)$
- ▶ The ATE estimand is characterized as an average over all clusters

$$\tau = \mathbb{E}[Y_{ij}(1) - Y_{ij}(0)]$$

- ▶ Under cluster randomization, can show

$$\tau = \mathbb{E}[Y_{ij}(1)|Z_i = 1] - \mathbb{E}[Y_{ij}(0)|Z_i = 0] = \mathbb{E}[Y_{ij}|Z_i = 1] - \mathbb{E}[Y_{ij}|Z_i = 0]$$

- ▶ The simple difference-in-means estimator is still consistent under cluster randomization

$$\hat{\tau}^{\text{unadj}} = \frac{\sum_{i=1}^n \sum_{j=1}^{m_i} Z_i Y_{ij}}{\sum_{i=1}^n Z_i m_i} - \frac{\sum_{i=1}^n \sum_{j=1}^{m_i} (1 - Z_i) Y_{ij}}{\sum_{i=1}^n (1 - Z_i) m_i}$$

Randomization-based Inference

(Gail et al. 1996)

- ▶ Randomization test can be carried out in the usual way to test the sharp null $H_0 : Y_{ij}(1) = Y_{ij}(0) \forall i, j$
- ▶ The unit of randomization is the cluster, and hence the randomization distribution should be created by permuting clusters (instead of individuals)
- ▶ Attractive because CRTs often recruit a few clusters (≤ 24), guarantee to control of test size at α level
- ▶ The weak null $H_0^* : \mathbb{E}[Y_{ij}(1) - Y_{ij}(0)] = 0$ is often more relevant
- ▶ Gail et al. (1996) showed that randomization test may have empirical test size exceeding the nominal level when H_0^* holds but H_0 violated

Randomization-based Inference

- ▶ A pair-matched design often substantially improve the power of CRT (Imai et al. 2009)
- ▶ When using pair-matched designs, the randomization test may have near nominal level even under the weak null
- ▶ Again the randomization test should respect the randomization design
 - ▶ in a pair-matched design, the possible number of randomization schemes is $2^{n/2}$
 - ▶ in a completely randomized design, the possible number of randomization schemes is $\binom{n}{n/2}$
 - ▶ only the former can be used to create the randomization distribution, otherwise the test may be invalid

Covariate Adjustment in CRT

- ▶ Both cluster-level and individual-level covariates can be collected in CRT
- ▶ CRT often has small to moderate number of clusters, covariate imbalance is common
- ▶ How to use regression to adjust covariate imbalance to gain efficiency? Regression at the cluster level or individual level?
- ▶ Su and Ding (2020): main conclusions
 - ▶ In large sample (in terms of number of clusters), regression of the **scaled cluster total outcome** on **scaled cluster total covariate and cluster size** is more efficient than the regression on individual data
 - ▶ Use the Huber-White (heteroskedasticity-robust) estimator of standard errors
 - ▶ Consistency and asymptotic normality holds even if the regression is misspecified

Covariate Adjustment: Additional Notes on Theory

- ▶ Two different ANCOVA models for covariate adjustment in RCT:

- ▶ ANCOVA1 (without interactions): $Y_i \sim \beta_0 + \beta_z Z_i + \beta_x^T X_i$

- ▶ ANCOVA2 (with interactions):

$$Y_i \sim \beta_0 + \beta_z Z_i + \beta_x^T (X_i - \bar{X}) + \beta_{xz}^T (X_i - \bar{X}) Z_i$$

- ▶ In both models, the OLS estimator of β_z is taken as the estimator of τ , often denoted as the adjusted estimator $\hat{\tau}^{adj}$
- ▶ Question: Any difference between the two OLS models? Which one is better?

Covariate Adjustment: General Theory

(Tsiatis et al. 2008)

- ▶ Recall ATE $\tau = \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)] = \mathbb{E}[Y|Z = 1] - \mathbb{E}[Y|Z = 0]$
- ▶ Sample size: $N_1 = \sum_{i=1}^N Z_i$, $N_0 = \sum_{i=1}^N (1 - Z_i)$, $N = N_1 + N_0$
- ▶ Randomization probability: $\bar{Z} = N^{-1} \sum_{i=1}^N Z_i = N_1/N$
- ▶ $\bar{Y}^{(1)} = N_1^{-1} \sum_{i=1}^N Z_i Y_i$, $\bar{Y}^{(0)} = N_0^{-1} \sum_{i=1}^N (1 - Z_i) Y_i$
- ▶ Can show all “reasonable” consistent and asymptotically normal estimators for either can be expressed exactly as or are asymptotically

$$\hat{\tau}_h = \bar{Y}^{(1)} - \bar{Y}^{(0)} - \sum_{i=1}^N (Z_i - \bar{Z}) \left\{ \frac{h^{(0)}(X_i)}{N_0} + \frac{h^{(1)}(X_i)}{N_1} \right\}$$

Covariate Adjustment: General Theory

(Tsiatis et al. 2008)

- ▶ General form of the ATE estimator

$$\hat{\tau}_h = \bar{Y}^{(1)} - \bar{Y}^{(0)} - \sum_{i=1}^N (Z_i - \bar{Z}) \left\{ \frac{h^{(0)}(X_i)}{N_0} + \frac{h^{(1)}(X_i)}{N_1} \right\}$$

- ▶ What happens if let $h^{(0)}(X_i) = h^{(1)}(X_i) \equiv 0$? (**Unadjusted!**)
- ▶ Consistency for arbitrary choice of h :
 - ▶ $\bar{Y}^{(1)} - \bar{Y}^{(0)}$ is obviously unbiased for τ
 - ▶ Invoke the Slutsky's theorem to show that the second piece $\xrightarrow{P} 0$;
intuitively this is because $\mathbb{E}(Z_i - \bar{Z}) = 0$
- ▶ Augmenting unadjusted estimator by the second term, which incorporates X and thereby implements the adjustment

Covariate Adjustment: ANCOVA1

- ▶ Model structure: $\mathbb{E}(Y_i|Z_i, X_i) = \beta_0 + \beta_z Z_i + \beta_x^T X_i$
- ▶ Can show that the least squares estimator $\hat{\beta}_z$ is

$$\left\{ 1 - \frac{1}{N_0 N_1} d_1^T \hat{\Sigma}_{XX}^{-1} d_1 \right\}^{-1} \left\{ \bar{Y}^{(1)} - \bar{Y}^{(0)} - \frac{N}{N_0 N_1} \sum_{i=1}^N (Z_i - \bar{Z}) \hat{\Sigma}_{XY}^T \hat{\Sigma}_{XX}^{-1} X_i \right\}$$

- ▶ $d_1 = \sum_{i=1}^N (Z_i - \bar{Z}) X_i$
- ▶ $\hat{\Sigma}_{XX} = N^{-1} \sum_{i=1}^N (X_i - \bar{X})(X_i - \bar{X})^T$ the sample covariance matrix of X
- ▶ $\hat{\Sigma}_{XY} = N^{-1} \sum_{i=1}^N (X_i - \bar{X})(Y_i - \bar{Y})$ sample covariance between X and Y

Covariate Adjustment: ANCOVA1

- ▶ OLS estimator $\hat{\beta}_z$ in ANCOVA1 is

$$\left\{ 1 - \frac{1}{N_0 N_1} d_1^T \hat{\Sigma}_{XX}^{-1} d_1 \right\}^{-1} \left\{ \bar{Y}^{(1)} - \bar{Y}^{(0)} - \frac{N}{N_0 N_1} \sum_{i=1}^N (Z_i - \bar{Z}) \hat{\Sigma}_{XY}^T \hat{\Sigma}_{XX}^{-1} X_i \right\}$$

- ▶ Weak Law of Large Numbers,

$$N^{-1} d_1 = N^{-1} \sum_{i=1}^N (Z_i - \bar{Z}) X_i \xrightarrow{P} 0$$

- ▶ $N^2/(N_0 N_1) \rightarrow \{\pi(1 - \pi)\}^{-1}$, with $N_1/N \rightarrow \pi = P(Z = 1)$

$$\frac{1}{N_0 N_1} d_1^T \hat{\Sigma}_{XX}^{-1} d_1 = \frac{N^2}{N_0 N_1} (N^{-1} d_1)^T \hat{\Sigma}_{XX}^{-1} (N^{-1} d_1) \xrightarrow{P} 0$$

- ▶ The first term simply converges to 1

Covariate Adjustment: ANCOVA1

Augmentation term

$$\frac{N}{N_0 N_1} \sum_{i=1}^N (Z_i - \bar{Z}) \hat{\Sigma}_{XY}^T \hat{\Sigma}_{XX}^{-1} X_i \approx \sum_{i=1}^N (Z_i - \bar{Z}) \left(\frac{1}{N_0} + \frac{1}{N_1} \right) \Sigma_{XY}^T \Sigma_{XX}^{-1} X_i$$

- ▶ $\Sigma_{XY} = \mathbb{E}[(X - \bar{X})(Y - \bar{Y})^T]$, $\Sigma_{XX} = \mathbb{E}[(X - \bar{X})(X - \bar{X})^T]$ true covariance matrices
- ▶ $\hat{\beta}_z$ corresponds to $\hat{\tau}_h$ with $h^{(0)}(X_i) = h^{(1)}(X_i) = \Sigma_{XY}^T \Sigma_{XX}^{-1} X_i$
- ▶ Regression coefficient estimator from ANCOVA1 is **consistent for ATE** and asymptotically normal **under entirely unrestrictive conditions**
- ▶ This is because the OLS estimator does not assume normality, constant variance or even the correctness of the model structure!!

Covariate Adjustment: ANCOVA2

- ▶ Model structure:

$$\mathbb{E}(Y_i|Z_i, X_i) = \beta_0 + \beta_z Z_i + \beta_x^T (X_i - \bar{X}) + \beta_{xz}^T (X_i - \bar{X}) Z_i$$

- ▶ Run OLS regression on centered covariates
- ▶ ANCOVA2 may seem inappropriate for estimating ATE τ , as the interaction implies that the conditional effect depends on the X and hence cannot equal τ
- ▶ But the OLS estimator $\hat{\beta}_z$ is again a special case of $\hat{\tau}_h$ and consistent to ATE **under entirely unrestrictive conditions**

Covariate Adjustment: ANCOVA2

- ▶ OLS estimator of β_z in ANCOVA2 is

$$\left\{ 1 - \frac{1}{N_0 N_1} d_2^T D^{-1} d_2 \right\}^{-1} \left\{ \bar{Y}^{(1)} - \bar{Y}^{(0)} - \frac{N}{N_0 N_1} d_2^T D^{-1} \begin{pmatrix} \hat{\Sigma}_{XY} \\ \hat{\Sigma}_{XYZ} \end{pmatrix} \right\}$$

- ▶ $d_2 = \{d_1^T, \sum_{i=1}^N (Z_i - \bar{Z})^2 (X_i - \bar{X})^T\}^T$
- ▶ $\hat{\Sigma}_{XYZ} = N^{-1} \sum_{i=1}^N (X_i - \bar{X})(Y_i - \bar{Y})(Z_i - \bar{Z})$
- ▶ $D = \begin{pmatrix} \hat{\Sigma}_{XX}^{(0)} & \hat{\Sigma}_{XX}^{(1)} \\ \hat{\Sigma}_{XX}^{(1)} & \hat{\Sigma}_{XX}^{(2)} \end{pmatrix}$ with

$$\hat{\Sigma}_{XX}^{(l)} = N^{-1} \sum_{i=1}^N (Z_i - \bar{Z})^l (X_i - \bar{X})(X_i - \bar{X})^T$$

- ▶ As $N^{-1} d_2 \xrightarrow{P} 0$, the first term (inside the inverse) converges to 1

Covariate Adjustment: ANCOVA2

Can show that augmentation term

$$\begin{aligned} & \frac{N}{N_0 N_1} d_2^T D^{-1} \begin{pmatrix} \hat{\Sigma}_{XY} \\ \hat{\Sigma}_{XYZ} \end{pmatrix} \\ & \approx \sum_{i=1}^N (Z_i - \bar{Z}) \left(\frac{1}{N_0} + \frac{1}{N_1} \right) \{ \pi \Sigma_{XY}^{(0)} + (1 - \pi) \Sigma_{XY}^{(1)} \}^T \Sigma_{XX}^{-1} X_i \end{aligned}$$

- ▶ $\Sigma_{XY}^{(l)} = \mathbb{E}[(X - \bar{X})(Y - \bar{Y})^T \mid Z = l]$, for $l = 0, 1$ is the conditional covariance
- ▶ This clearly shows that $\hat{\beta}_Z$ corresponds to $\hat{\tau}_h$ with $h^{(0)}(X_i) = h^{(1)}(X_i) = \{ \pi \Sigma_{XY}^{(0)} + (1 - \pi) \Sigma_{XY}^{(1)} \}^T \Sigma_{XX}^{-1} X_i$
- ▶ ANCOVA2 is consistent for τ and asymptotically **even if the model structure is an incorrect representation of $\mathbb{E}(Y|Z, X)$**

Comparing ANCOVA1 and ANCOVA2

- ▶ Regression coefficient both consistent to ATE
- ▶ Can write the augmentation term of ANCOVA1,

$$h^{(0)}(X_i) = h^{(1)}(X_i) = \Sigma_{XY}^T \Sigma_{XX}^{-1} X_i = \{(1 - \pi) \Sigma_{XY}^{(0)} + \pi \Sigma_{XY}^{(1)}\}^T \Sigma_{XX}^{-1} X_i$$

- ▶ Compare with the augmentation term in ANCOVA2,

$$h^{(0)}(X_i) = h^{(1)}(X_i) = \{\pi \Sigma_{XY}^{(0)} + (1 - \pi) \Sigma_{XY}^{(1)}\}^T \Sigma_{XX}^{-1} X_i$$

- ▶ ANCOVA1 and ANCOVA2 are asymptotically **equivalent** equal randomization $\pi = 1/2$!

Comparing ANCOVA1 and ANCOVA2

- ▶ For general cases, ANCOVA2 has smaller asymptotic variance than ANCOVA1
- ▶ By appealing to the semiparametric efficiency theory, ANCOVA2 estimator $\hat{\beta}_z$ has the smallest asymptotic variance among the class of estimators $\hat{\tau}_h$ for which $h^{(l)}(X_i)$ are linear functions of X_i
- ▶ More interestingly, ANCOVA2 is asymptotically equivalent to propensity score weighting estimators (later)
- ▶ Cautionary note is that all of these interesting results are established under linear models

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